Big Data in Public Safety: Enhancing Law Enforcement Options

Guided by: Professor Anthony Fantasia

Image reference: https://images.app.goo.gl/exLrLWJMiD2p3RB67

Presented by:
Esther, Gauri, Nisha, Pardhiv, Pranav & Srilakshmi

Group – 4
ADTA 5240

Public Sector | Security Use Case

- Police forces are increasingly using big data tools to monitor suspicious criminal activities, predict crime hotspots, and improve resource allocation.

- Our project demonstrates how the Dallas Police Department can use arrest data analytics for smarter policing.

Data Lifecycle

01 Generation

02 Collection

03 Processing

04 Storage

05 Management

06 Analysis

07 Visualization

08 Interpretation

Generation

- Data sourced from the Dallas Open Data Portal.
- The source contains information about the arrests happening in the area of Dallas.
- The data is updated everyday with latest arrest data.
- Helps identify crime patterns and trends across the city.

Source: https://www.dallasopendata.com/Public-Safety/Police-Arrests/sdr7-6v3j/about_data

Collection

- The data extracted on April 12th is considered as a static dataset for this project.
- The source data had 65 columns, only 33 of those columns were chosen for storage and processing.
- Used Socrata API and to automate data extraction.
- Initial raw data included inconsistencies and missing fields.

Processing

Used **Google Cloud OpenRefine** for data cleaning:

- Duplicate removal
- Columns with personal data and irrelevant data were eliminated.
- Only the columns contributing to the problem statement were choosen.
- Used text clustering to correct spelling errors and unify inconsistent entries.
- Replaced items marked as “Unknown” with NULL.
- Aligned data types with corresponding fields (e.g., string, integer, date).
- Convert height (e.g., 5-11 → 5.11). So that it is in Float type.

OpenRefine

Data Storage

Google BigQuery:

Used for real-time data access and querying through SQL-like BigQuery syntax.

Apache Spark with HDFS:

Used for distributed data storage and large-scale processing.

Batch Processing Enabled:

Python scripts integrated with both platforms to support ad-hoc batch processing tasks.

Image Reference: https://www.techrepublic.com/article/google-cloud-platform-the-smart-persons-guide/

Data Storage

Google BigQuery

- Auto Schema detection feature was used to create a schema table that represents the actual data.

Apache Spark and Hive with HDFS

- Data is uploaded to a HDFS folder and a table with the same schema is created.

```
INFO  : Concurrency mode is disabled, not creating a lock manager
INFO  : Executing command(queryId=hive_20250420125653_3d21ceaf-69a2-4cfa-9861-67c86c7b30a1): CREATE EXTERNAL TABLE IF NOT EXISTS arrest_data_dallas
(
`age` INT,
`occupation` STRING,
`drugrelated` BOOLEAN,
`tatoo` STRING,
`birthplace` STRING,
`drugtype` STRING,
`drug` BOOLEAN,
`sex` STRING,
`height` DOUBLE,
`nickname` BOOLEAN,
`arcurrloc` STRING,
`hzip` INT,
`upzdate` TIMESTAMP,
`araction` STRING,
`employer` STRING,
`arpremises` STRING,
`aradow` STRING,
`arldistrict` STRING,
`arrestyr` INT,
`hstate` STRING,
`hcity` STRING,
`ethnic` STRING,
`race` STRING,
`eyes` STRING,
`hair` STRING,
`ageatarresttime` INT,
`arweapon` STRING,
`arlcounty` STRING,
`arstate` STRING,
`arlcity` STRING,
`arlzip` INT,
`arladdress` STRING,
`arbkdate` TIMESTAMP,
`ararresttime` STRING,
`ararrestdate` TIMESTAMP,
`weight` INT
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
STORED AS TEXTFILE
LOCATION '/user/pranavmoses/data/arrestdata/'
TBLPROPERTIES ("skip.header.line.count"="1")
INFO  : Starting task [Stage-0:DDL] in serial mode
INFO  : Completed executing command(queryId=hive_20250420125653_3d21ceaf-69a2-4cfa-9861-67c86c7b30a1); Time taken: 0.126 seconds
INFO  : OK
INFO  : Concurrency mode is disabled, not creating a lock manager
No rows affected (0.185 seconds)
```

Schema details in Apache eco system

Repositories
Queries
Notebooks
Data canvases
Data preparations
Pipelines
External connections
adta_5240_project
dallas_arrest_data

Repository Preview

No repository selected

Filter Enter property name or value

Field name	Type	Mode
age	INTEGER	NULLABLE
occupation	STRING	NULLABLE
drugrelated	BOOLEAN	NULLABLE
tatoo	STRING	NULLABLE
birthplace	STRING	NULLABLE
drugtype	STRING	NULLABLE
drug	BOOLEAN	NULLABLE
sex	STRING	NULLABLE
height	FLOAT	NULLABLE
nickname	BOOLEAN	NULLABLE
arcurrloc	STRING	NULLABLE
hzip	INTEGER	NULLABLE
upzdate	TIMESTAMP	NULLABLE
araction	STRING	NULLABLE
employer	STRING	NULLABLE
arpremises	STRING	NULLABLE
aradow	STRING	NULLABLE
arldistrict	STRING	NULLABLE
arrestyr	INTEGER	NULLABLE
hstate	STRING	NULLABLE
hcity	STRING	NULLABLE
ethnic	STRING	NULLABLE
race	STRING	NULLABLE
eyes	STRING	NULLABLE
hair	STRING	NULLABLE
ageatarresttime	INTEGER	NULLABLE
arweapon	STRING	NULLABLE
arlcounty	STRING	NULLABLE
arstate	STRING	NULLABLE
arlcity	STRING	NULLABLE
arlzip	INTEGER	NULLABLE
arladdress	STRING	NULLABLE
arbkdate	TIMESTAMP	NULLABLE
ararresttime	STRING	NULLABLE
ararrestdate	TIMESTAMP	NULLABLE
weight	INTEGER	NULLABLE

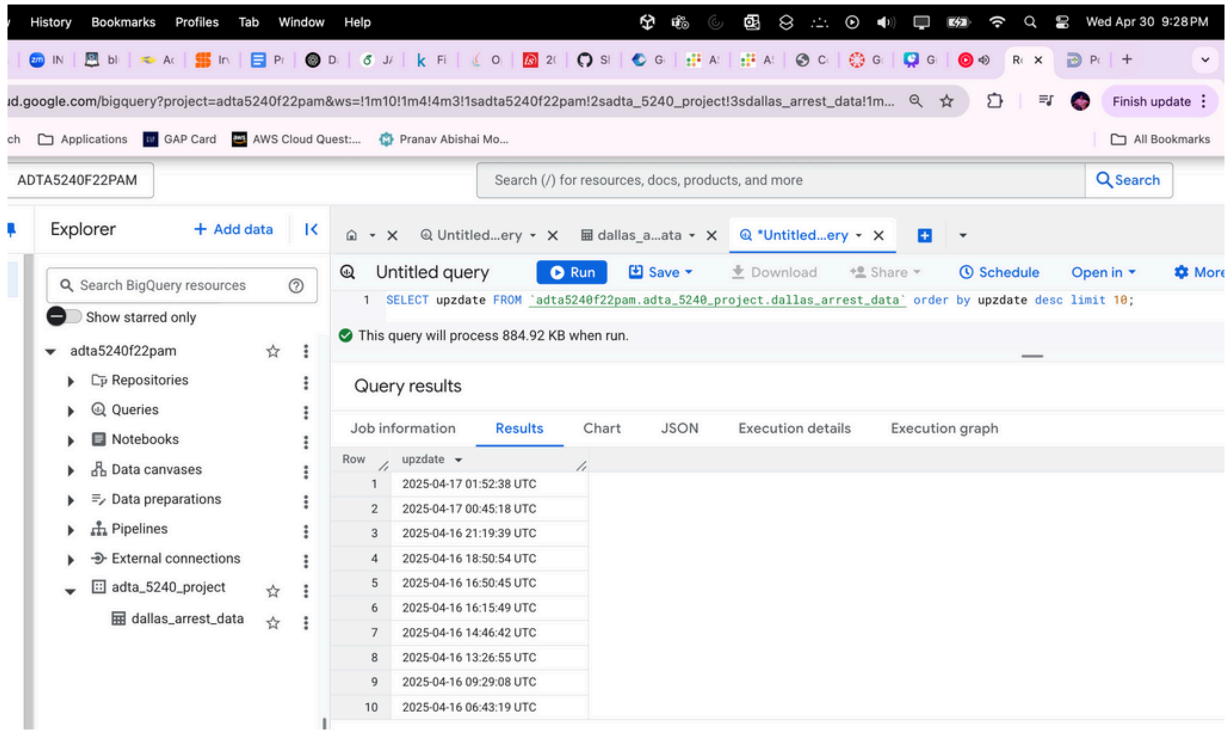
Schema tab in BigQuery

Image Reference: Project and Dataset

Data Storage - Streaming Data

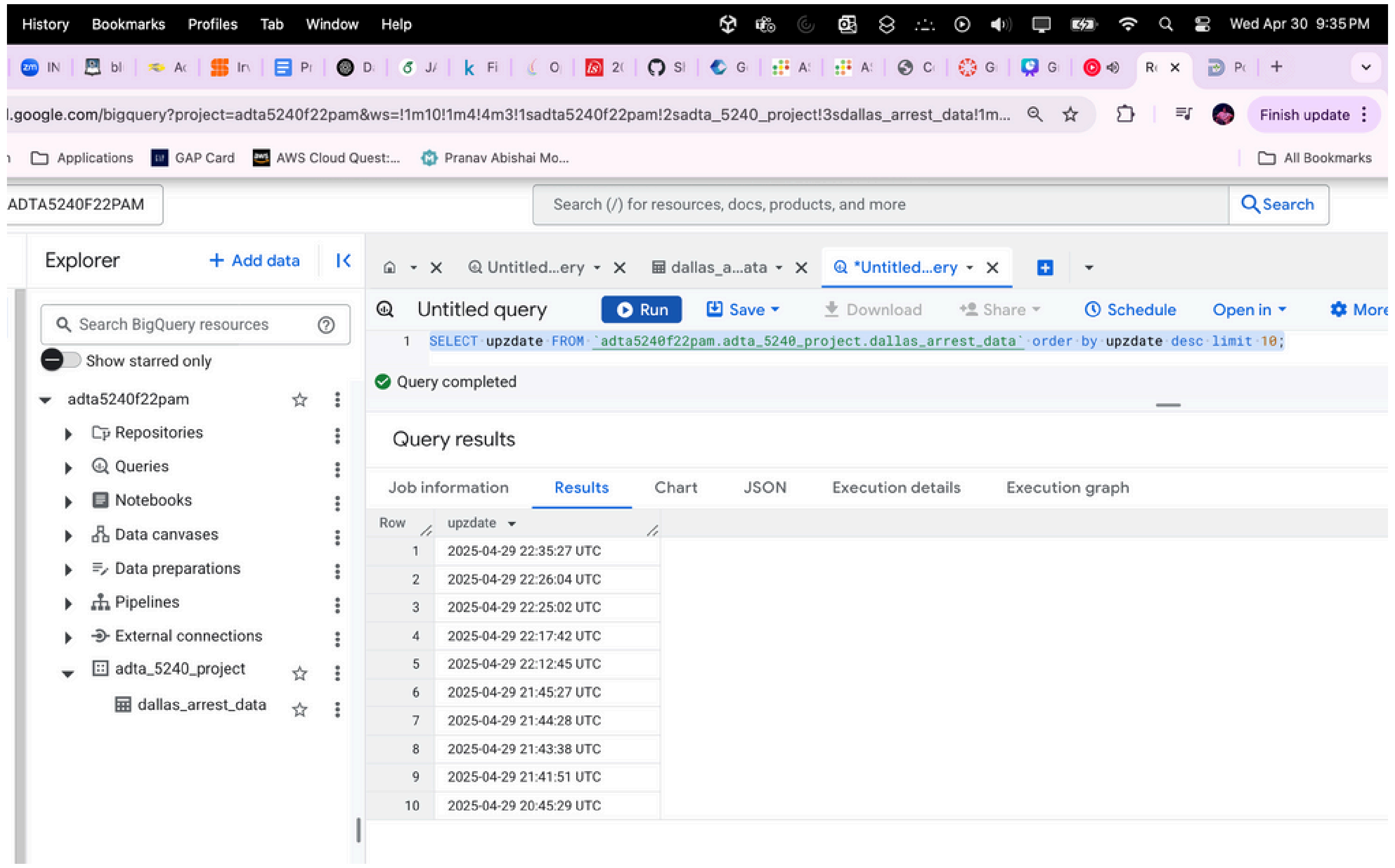
Google BigQuery

The column “upzdate” was used to extract new data from the data source.



Before running the python script

Python script runs successfully



After running the python script

Image Reference: Project and Dataset

Data Storage - Streaming Data

Apache HDFS EcoSystem

The column “upzdate” was used to extract new data from the data source.

```
spark-sql> select upzdate from arrest_data_dallas order by upzdate desc limit 5;
25/05/01 02:47:50 WARN org.apache.hadoop.hive.ql.session.SessionState: METASTORE_
ctory.
2025-04-19 20:24:25
2025-04-19 19:20:44
2025-04-19 18:16:27
2025-04-19 17:51:55
2025-04-19 17:38:39
Time taken: 8.803 seconds, Fetched 5 row(s)
spark-sql>
```

Before running the python script

```
25/05/01 02:54:57 WARN org.apache.spark.sql.catalyst.util.package: Truncated the string representation
ebug.maxToStringFields'.
25/05/01 02:54:58 INFO org.apache.hadoop.mapred.FileInputFormat: Total input files to process : 7
Maximum upzdate in table: 2025-04-19 20:24:25
WARNING:root:Requests made without an app_token will be subject to strict throttling limits.
25/05/01 02:55:06 INFO org.apache.hadoop.hive.common.FileUtils: Creating directory if it doesn't exist:
hive_2025-05-01_02-55-06_603_2754448413448O2284-1
25/05/01 02:55:15 INFO org.apache.hadoop.hive.ql.security.authorization.plugin.sqlstd.SQLStdHiveAccessC
text [sessionString=d1b45ea6-4de1-4948-a973-75bb3fd39f7a, clientType=HIVECLI]
25/05/01 02:55:15 WARN org.apache.hadoop.hive.ql.session.SessionState: METASTORE_FILTER_HOOK will be ig
ctory.
25/05/01 02:55:15 INFO hive.metastore: Mestastore configuration hive.metastore.filter.hook changed from
ive.ql.security.authorization.plugin.AuthorizationMetaStoreFilterHook
25/05/01 02:55:15 INFO hive.metastore: Closed a connection to metastore, current connections: 0
25/05/01 02:55:15 INFO hive.metastore: Trying to connect to metastore with URI thrift://dp-hadoop-spark
25/05/01 02:55:15 INFO hive.metastore: Opened a connection to metastore, current connections: 1
25/05/01 02:55:15 INFO hive.metastore: Connected to metastore.
25/05/01 02:55:15 INFO hive.metastore: Trying to connect to metastore with URI thrift://dp-hadoop-spark
25/05/01 02:55:15 INFO hive.metastore: Opened a connection to metastore, current connections: 2
25/05/01 02:55:15 INFO hive.metastore: Connected to metastore.
Number of rows updated: 321
```

Python script runs successfully

```
Time taken: 8.803 seconds, Fetched 5 row(s)
spark-sql> select upzdate from arrest_data_dallas order by upzdate desc limit 5;
2025-04-29 22:35:27
2025-04-29 22:25:02
2025-04-29 22:17:42
2025-04-29 22:12:45
2025-04-29 21:45:27
Time taken: 6.067 seconds, Fetched 5 row(s)
spark-sql>
```

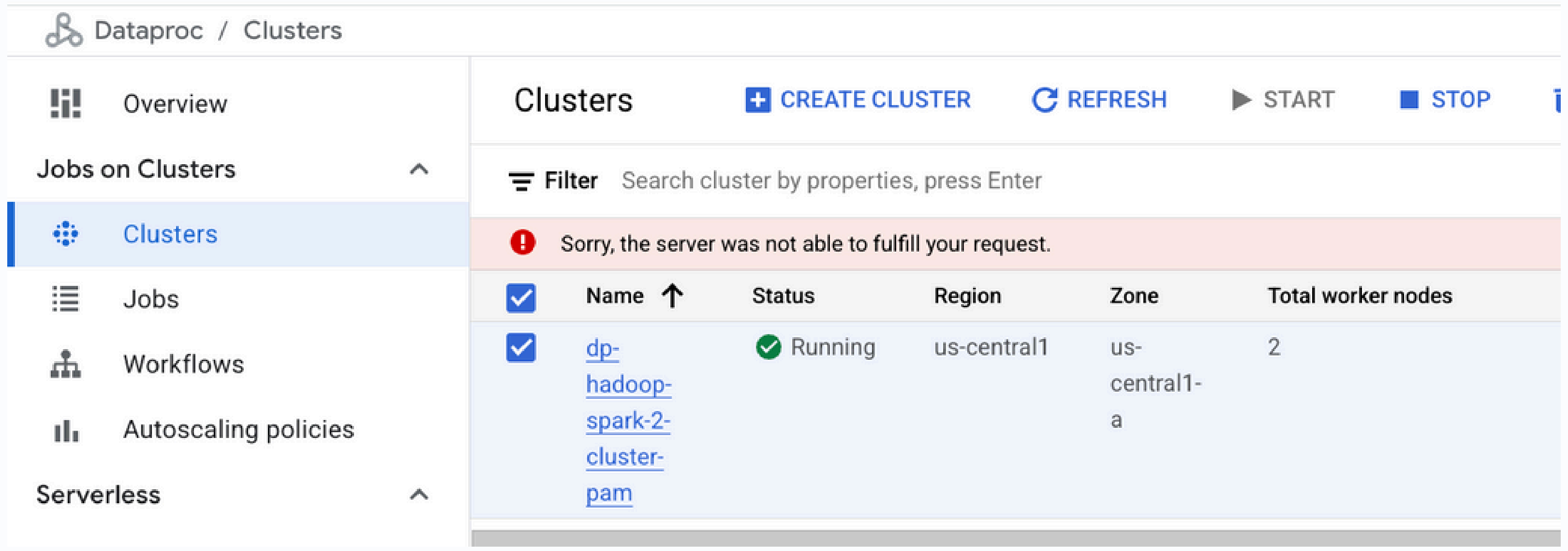
After running the python script

Image Reference: Project and Dataset

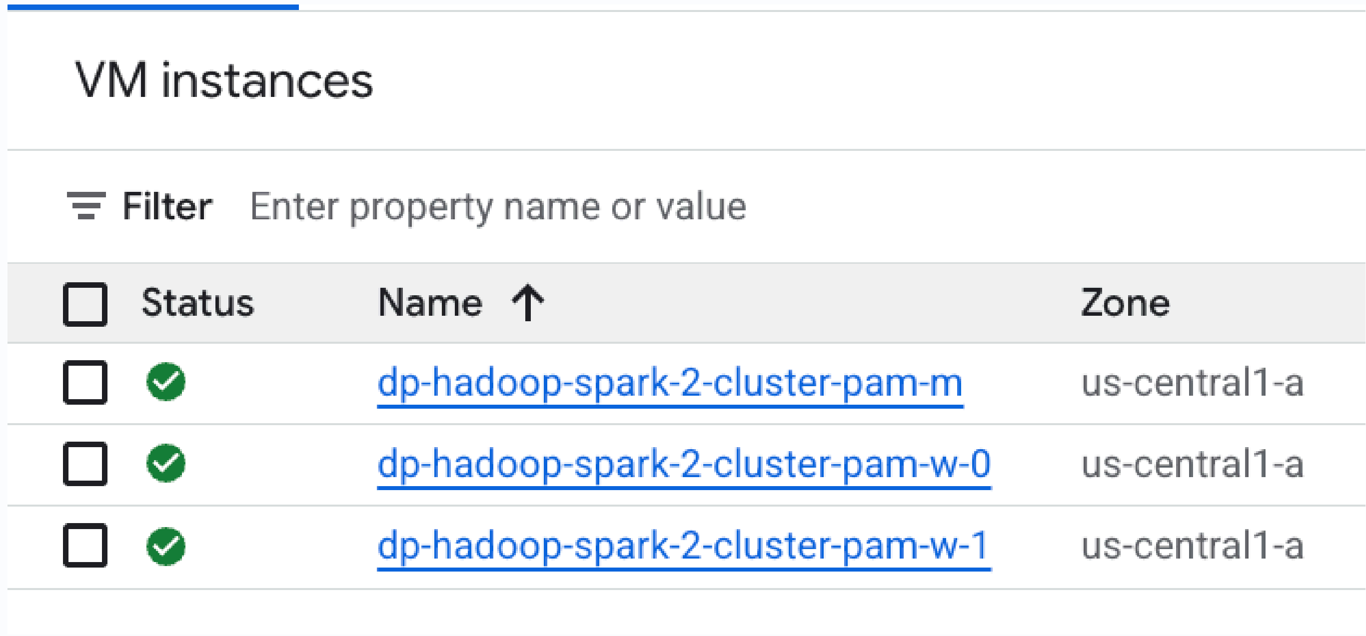
Management

The data collected is organized using Google OpenRefine and stored on cloud in: BigQuery and Hadoop EcoSystems

Dataproc Cluster is created with a namenode and two worker nodes to process data in the Hadoop EcoSystem.



Dataproc cluster

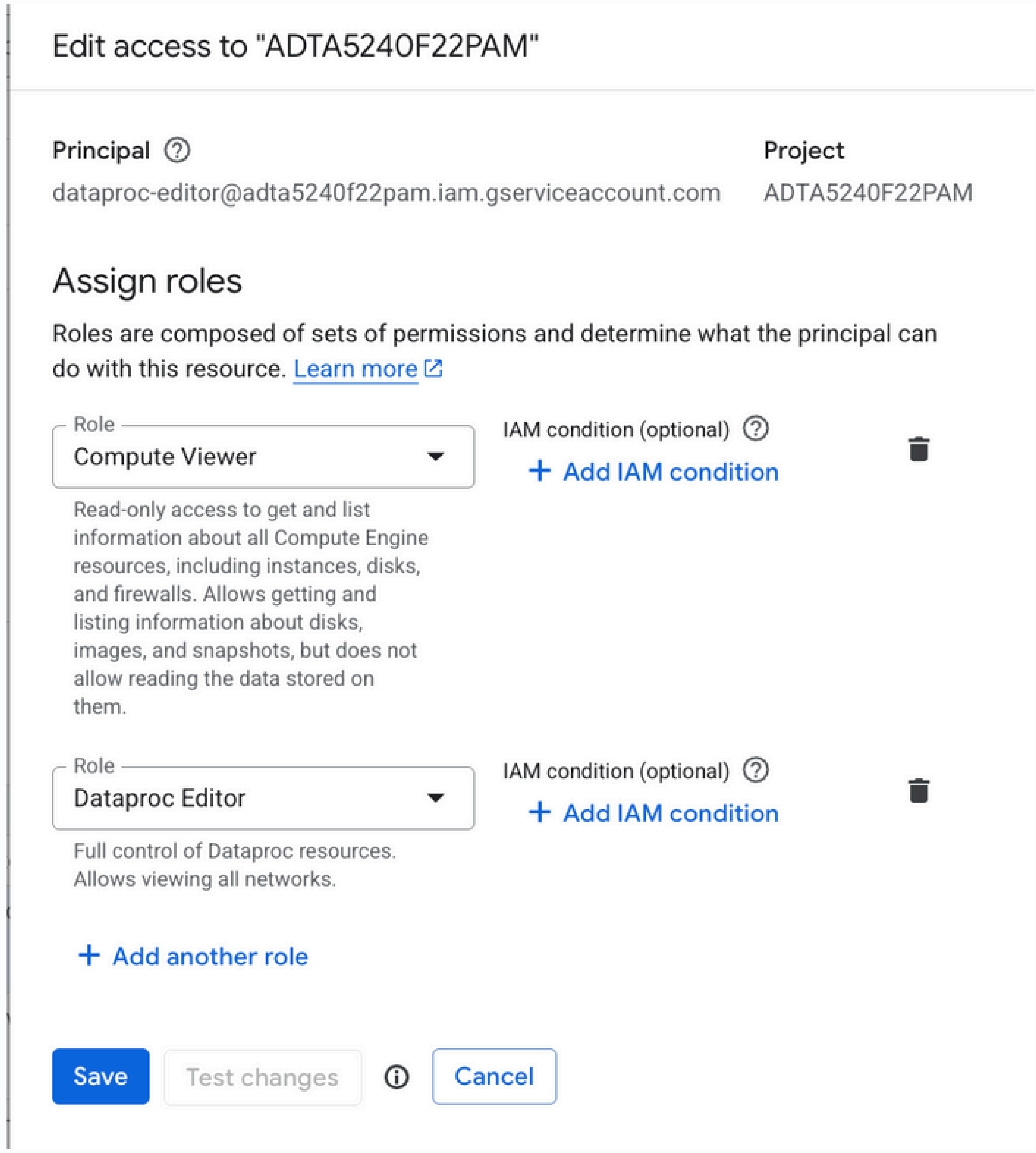


VM instances created for the Dataproc Cluster

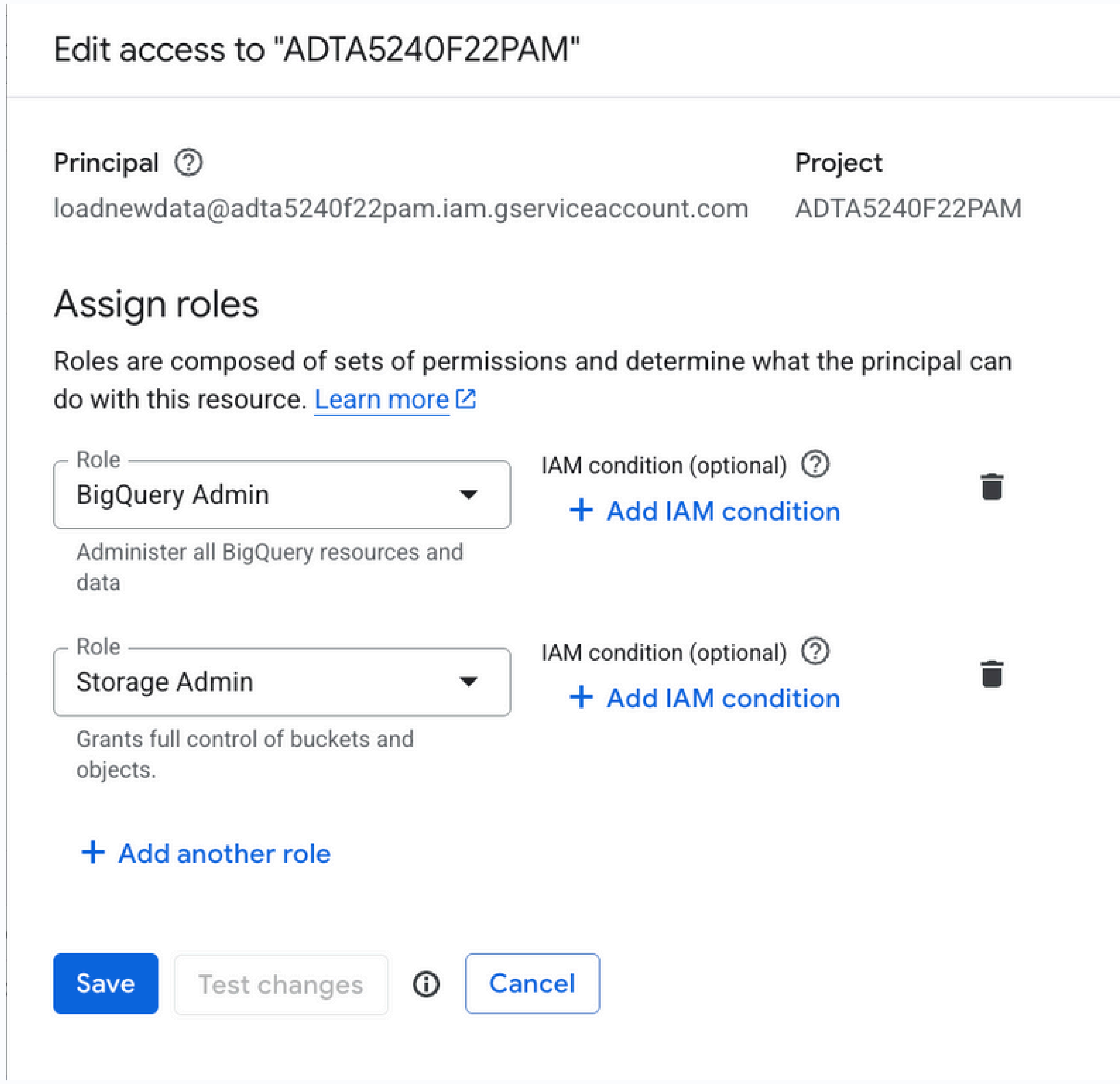
Image Reference: Project and Dataset

Management

Access is given to a service account and scripts are triggered using the same service account.



Dataproc Editor access service account



BigQuery and Storage admin access service account

Image Reference: Project and Dataset

Analysis

Query results

Job information | Results | Chart | JSON | Execution details | Execut

Row	Sex	ArCurrLoc	average_age
1	MALE	DX	37.0
2	MALE	LS	35.0
3	FEMALE	LS	34.0
4	FEMALE	DX	33.0
5	TEST	LS	26.0

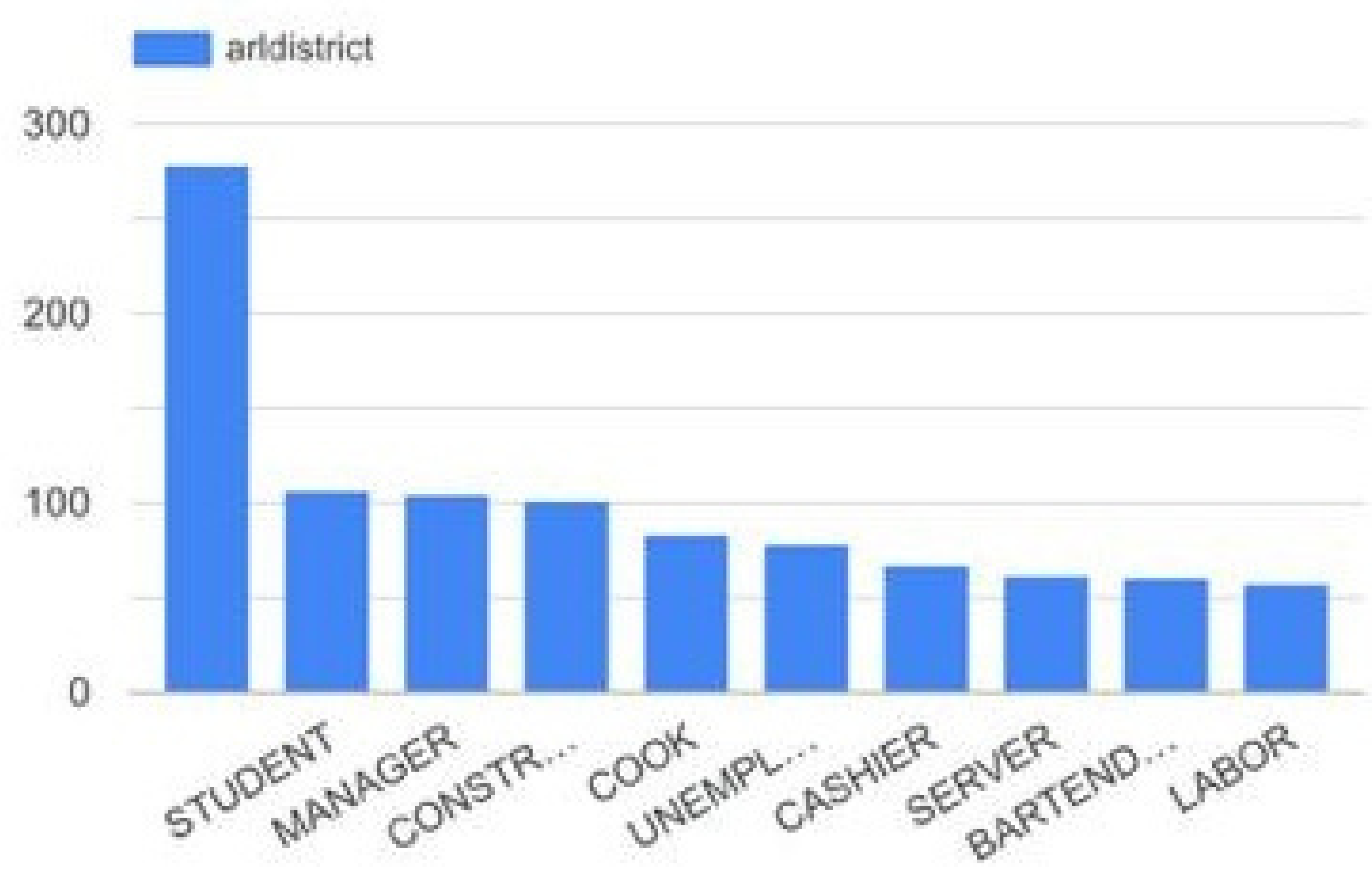
Analysis

- Key Insights
  - Males in DX have the highest average arrest age (35.37 years), suggesting older individuals are more frequently arrested there.
  - Female arrestees show a consistent average age across both locations (31years).
  - Black individuals represent the highest arrest counts in both LS (34,700) and DX (13,244), followed by Hispanic or Latino.
  - LS consistently has more arrests than DX for all racial groups.
  - DX recorded the highest single-day arrests with 31 arrests on 2014-06-01, indicating a possible concentrated policing event or high-crime day.
  - Midnight (00:00) is the most common arrest time, with 1,163 arrests, indicating heightened law enforcement activity or incidents right at the start of the day.

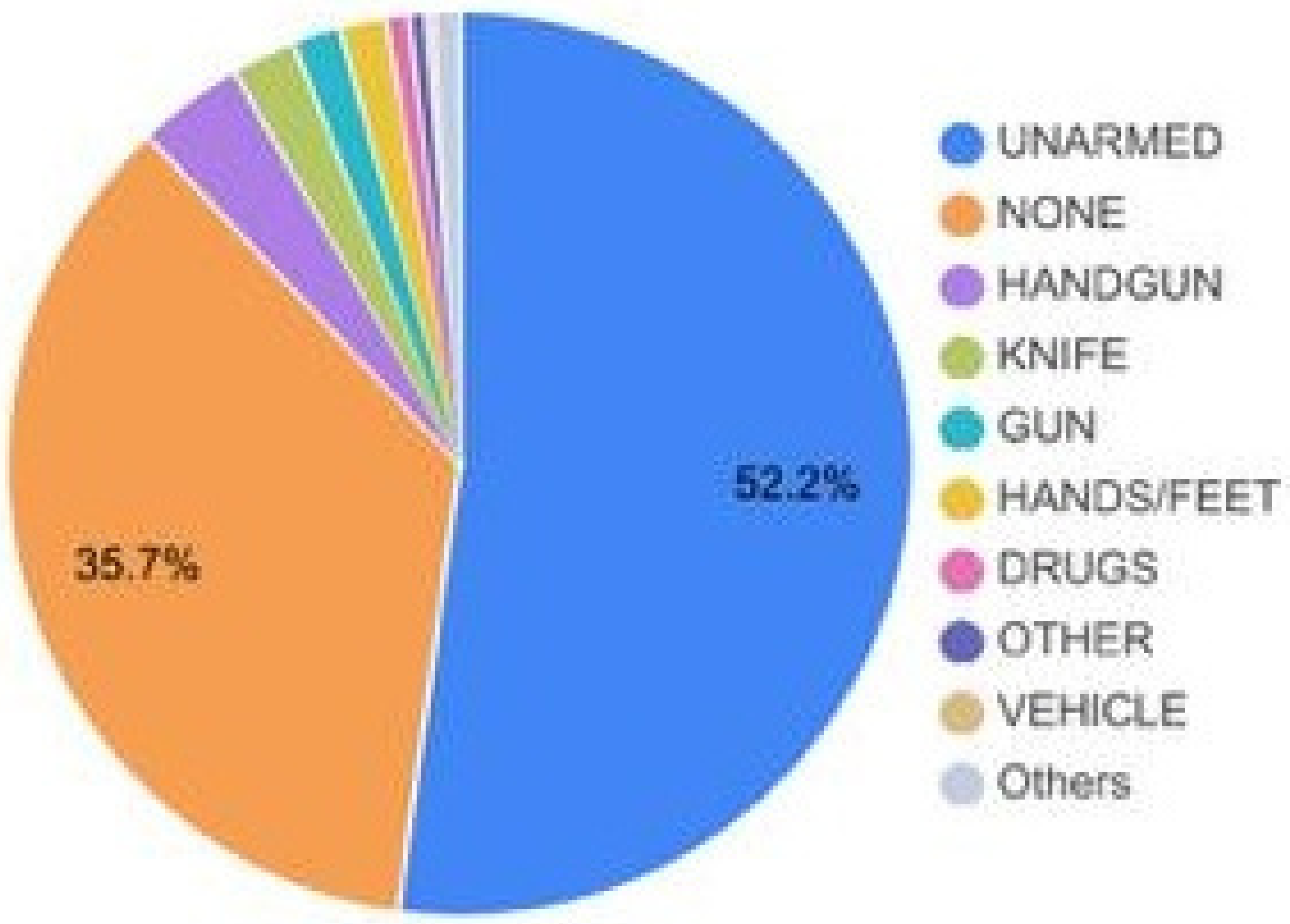
Visualization

- Built using Google Looker Studio & Tableau.
- Showed trends by race, year, offense type, and weapon involvement.
- Visuals made patterns clear for quick insights.
- Helped identify hotspots and disparities for better planning.

Visualization



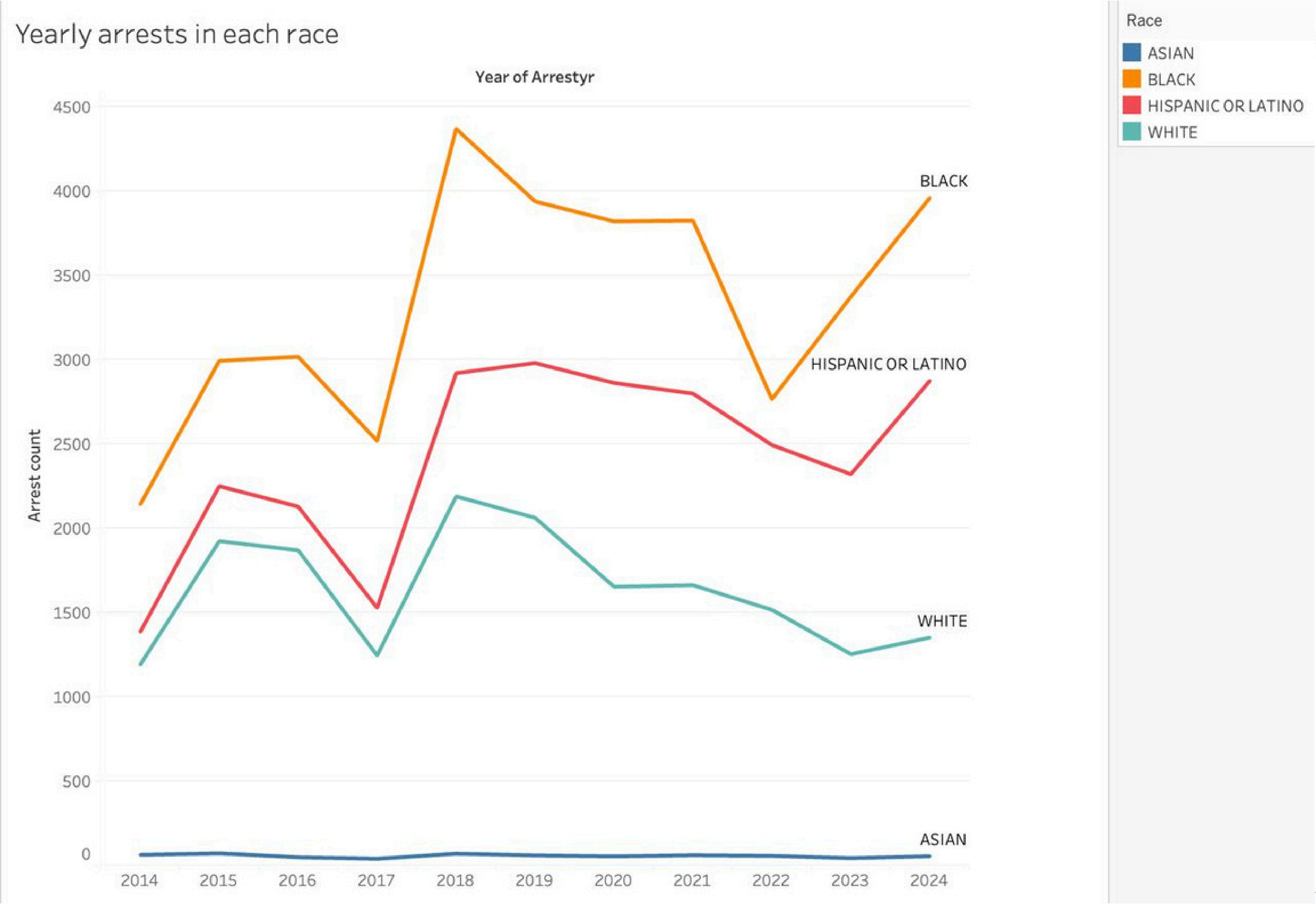
Arrests by Employment Type



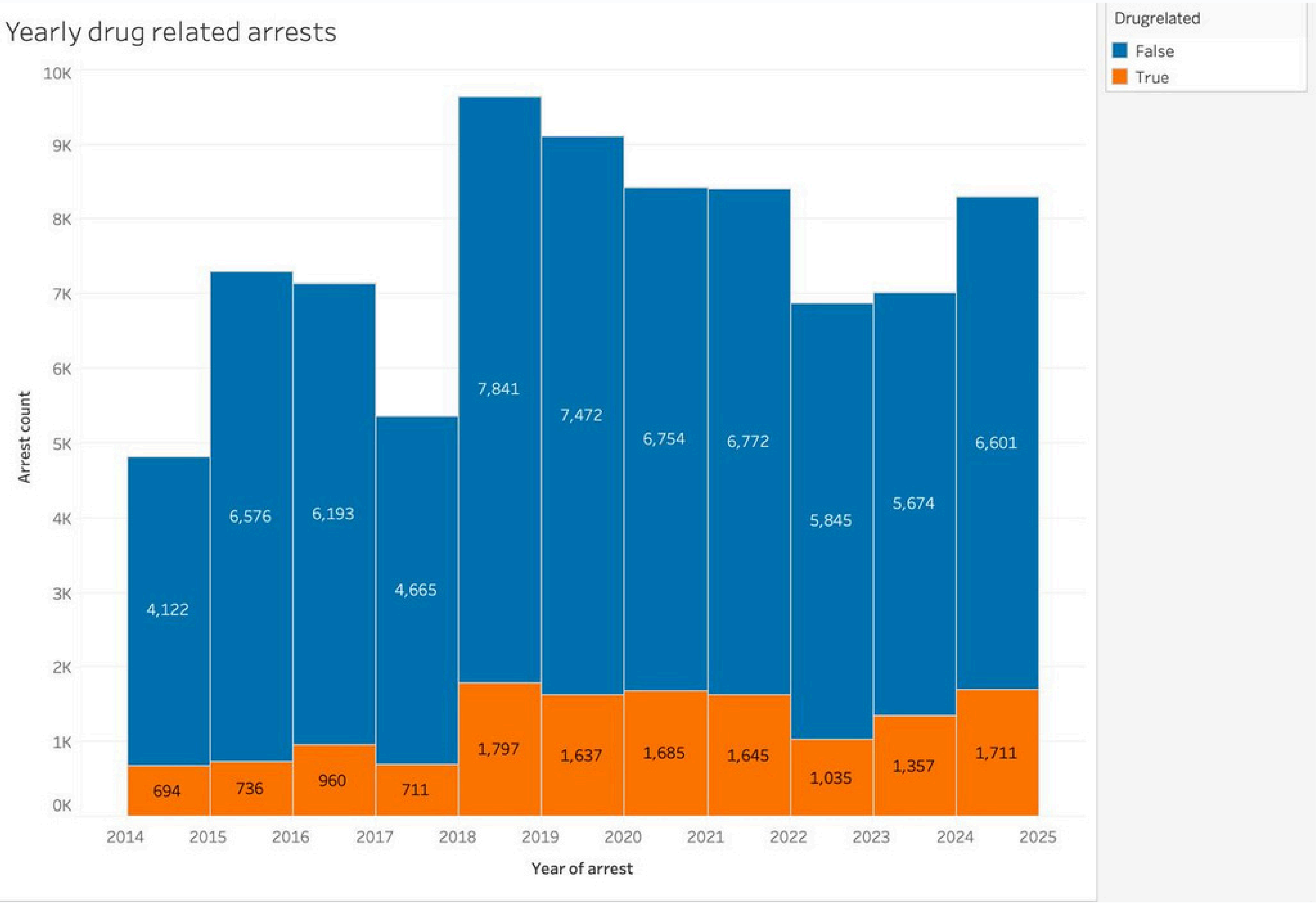
Object Involved at Time of Arrest

Image Reference: Project and Dataset - Looker Studio

Visualization



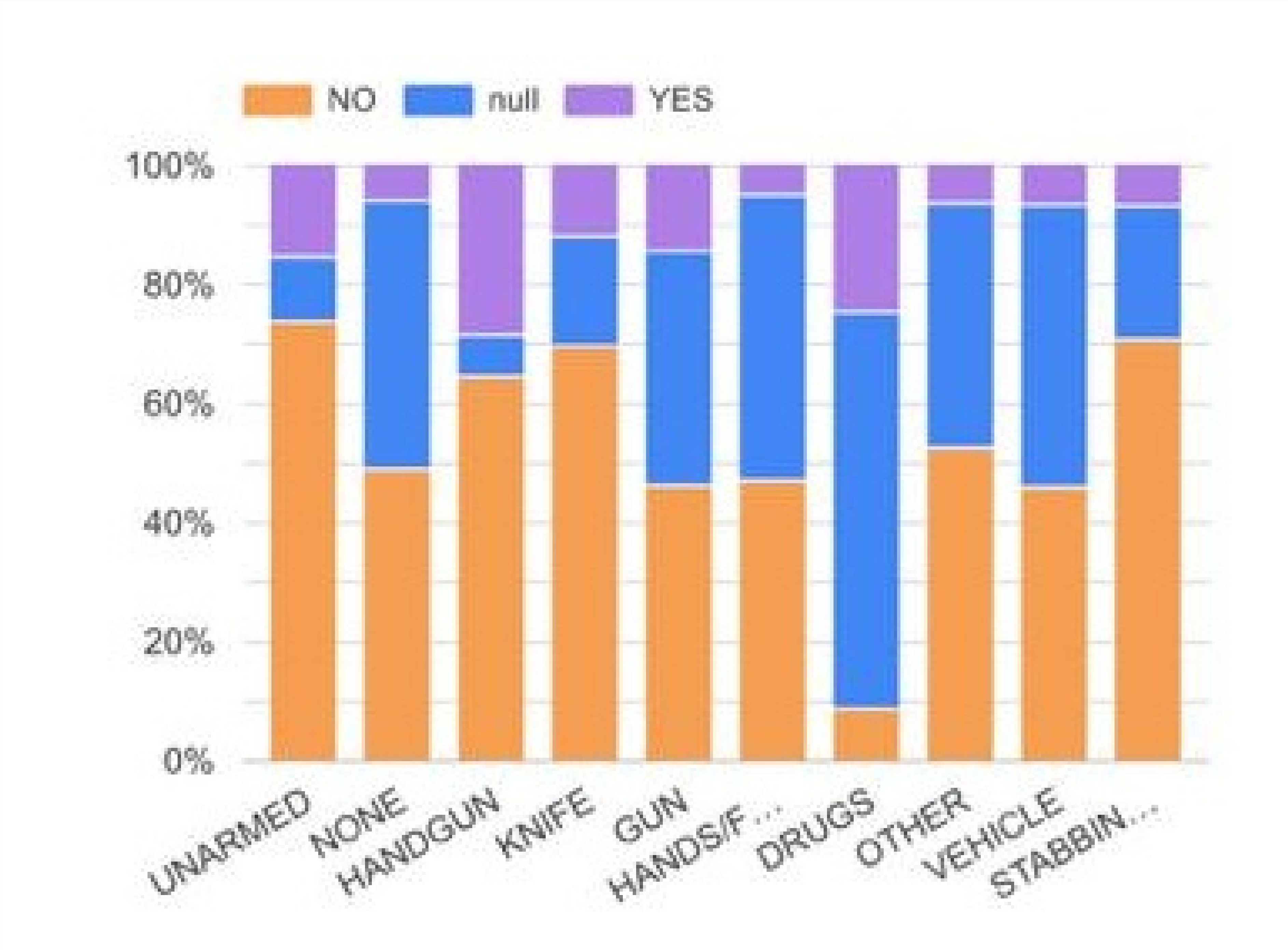
Trends in Yearly Arrests by Race (2014–2024)



Drug-Related vs. Non-Drug Arrests (2014–2025)

Image Reference: Project and Dataset - Tableau

Visualization



Use of Force by Object Involved

Image Reference: Project and Dataset - Looker Studio

Interpretation

- Arrest trends show systemic disparities across race and employment type.
- Drug enforcement remains a key driver of arrest activity.
- Unarmed individuals frequently invloved in use-of-force incidents.
- Visual insights help uncover patterns not obvious in raw data.
- Supports informed decisions in policy, training, and community policing.

Recommendations

- Audit racial trends to address potential bias.
- Expand public healt-based responses to drug arrests.
- Launch outreach for students and low-wage workers.
- Improve de-escalation and use-of-force training.
- Share public dashboards for transparency and trust.

Conclusion

- Arrest data reveals critical areas for police reform.
- Analytics tools support smarter, fairer policing.
- Data-driven stratgies lead to safer communities.
- Open data strategies lead to safer commuities.
- Open data + big data = better decisions and accountability.

Thank You

Any Questions?

Image reference: https://images.app.goo.gl/dYf3yiVzaSpYZEKM7

Group – 4
ADTA 5240